Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Parallel axis

Meaning in my words How would you explain parallel axis to a friend?	**How can you tell?** If downhill is positive, what sign does the along-ramp part of gravity have?
Example or sketch Sketch axes for a ramp and mark the parallel axis.	**Non-example and why** A partner uses the horizontal floor as the parallel axis. Which surface should they follow?

Perpendicular axis

Meaning in my words How would you explain perpendicular axis to a friend?	**How can you tell?** If the positive axis points away from the ramp, what sign does the normal force have?
Example or sketch Draw the perpendicular axis for a block on a tilted ramp.	**Non-example and why** A student draws that axis straight up on a tilted ramp. Why would you rotate it?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Weight components

Meaning in my words What are the two components of weight on a ramp?	**How can you tell?** Which part of weight pulls downhill, and which presses into the ramp?
Example or sketch Draw weight and its two parts. Which arrows should enter the two ramp-axis equations?	**Non-example and why** A student adds weight and both of its components as three forces. What got counted twice?

Constraint

Meaning in my words How would you explain constraint to a friend?	**How can you tell?** A block stays on a straight ramp. What is its acceleration perpendicular to the surface?
Example or sketch How would you use that fact to find the normal force when gravity and the surface are the only forces?	**Non-example and why** Does zero perpendicular acceleration mean zero downhill acceleration?

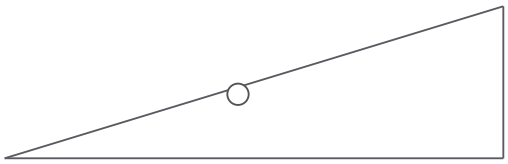
Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER

Name _____

Draw the real weight once, then draw its two components in a separate box. Circle the component along the ramp.

Real forces



Components

My marks mean

On assignment: Weight components

SECTION E / BACK / FINAL MEETING ONLY Name _____

A navigation graphic lists a gravitational force AND its two components as three separate pulls on XJ.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Navigation engineer**

Build an overlay using the component view. Indicate the full arrow it replaces before releasing the values to the calculator.

☐ **Map graphics editor**

Sketch two camera views: the real pull, then its components. Give viewers a caption that prevents counting the same pull twice.

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

Editor's check: State which component can change speed at that instant.